Journal of Analytical and Applied Pyrolysis 196 (2026) 107733

Contents lists available at ScienceDirect

Journal of Analytical and Applied Pyrolysis

journal homepage: www.elsevier.com/locate/jaap

Pyrolysis mechanism of engineering plastic waste under carbon dioxide

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ARTICLE INFO

Keywords:
Thermochemical conversion
Polybutylene terephthalate
Carbon dioxide
Syngas
Carbon-negative process

ABSTRACT

This study explored carbon CO_2-assisted catalytic pyrolysis as a sustainable method for converting polybutylene terephthalate (PBT) waste into syngas under moderate-temperature conditions (≤700 ℃), providing a more energy-efficient alternative to conventional gasification. CO_2 was adopted as a reactive medium to enhance gas-phase carbon conversion and reduce the environmental footprint. The results showed that CO_2-assisted catalytic pyrolysis effectively suppressed the formation of condensable pyrogenic products, reducing the liquid yields from 67.91 wt% to 10 wt%, and boosted CO production by up to 65-fold compared to that of one-stage pyrolysis under inert conditions. To elucidate the fundamental reaction pathways, terephthalic acid (TPA) was utilized as a model compound representing the core aromatic backbone of PBT. Mechanistic analysis showed that CO_2 activation on the Ni catalyst surface generated reactive intermediates ($CO_2{\bullet}^-$ and M-COO$^-$), which promoted selective bond cleavage and deoxygenation reactions, efficiently converting heavy oxygenates into syngas. Moreover, CO_2 played a dual role by also suppressing coke formation, thereby preserving the catalyst performance. An environmental assessment indicated that the catalytic pyrolysis of 1.8 million tonnes of PBT waste could lead to a net reduction of 6 million tonnes of CO_2 emissions compared to that under incineration, which would release nearly 4 million tonnes. These findings highlight the use of CO_2-assisted catalytic pyrolysis as a carbon-negative and circular solution that transforms plastic waste and CO_2 into valuable products.

1. Introduction

Plastics are becoming increasingly important materials in the automotive industry because of their contribution to improving fuel efficiency through vehicle weight reduction [1,2]. The plastic content of an average vehicle was 6 wt% in 1970, which increased to 16 wt% in 2010 and is expected to reach 30 wt% by 2030 [1]. In addition, more than 13 different types of plastics are used in vehicle manufacturing. These include polypropylene, polyvinyl chloride, polyurethane, polyamide, acrylonitrile butadiene styrene, polycarbonate, high-density polyethylene, and polybutylene terephthalate (PBT) [3]. The use of such materials has resulted in a 200-kg reduction in the average weight of vehicles [4]. However, the increasing reliance on plastics has shifted environmental burdens from the operational phase (*e.g.*, reduced greenhouse gas emissions) to the end-of-life vehicle (ELVs) stage, where disposal issues have become more considerable. Landfilling is currently the predominant method of managing automobile plastic waste [5]. The degradation of plastic components in landfills leads to the release of microplastics, which accumulate in leachate [6,7] and may subsequently disperse into the surrounding environment [8]. These microplastics persist in nature and are taken up by organisms, potentially causing ecological harm [9]. In response to these concerns, incineration has been widely employed as an alternative to landfilling [10]. Although it potentially offers benefits such as energy recovery and volume reduction, plastic incineration is often limited in terms of its ability to achieve complete oxidation. This leads to the formation of toxic compounds, including unburned hydrocarbons (HCs), polycyclic aromatic HCs (PAHs), dioxins, and furan analogs [11,12]. Therefore, developing environmentally benign strategies for managing automobile plastic waste is essential for sustainably minimizing environmental burdens.

Plastic waste generated from automobiles should not be regarded merely as environmental pollutants; rather, they should be considered

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https://doi.org/10.1016/j.jaap.2026.107733
Received 17 November 2025; Received in revised form 18 February 2026; Accepted 5 March 2026
Available online 7 March 2026

valuable resources. Their HC-rich structure renders them suitable as potential feedstocks for producing fuels and chemical intermediates [13]. Consequently, recycling can mitigate environmental burdens and aligns with circular economy principles by extending the lifecycle of materials and enhancing resource efficiency [14]. However, the practical implementation of recycling practices for automobile plastic waste remains limited [15]. This is mainly because of the heterogeneous and complex composition of such waste streams, which often include multiple polymer types, additives, and reinforcements (carbon fibers) that are physically and chemically interconnected [15,16]. These characteristics make mechanical recycling and chemical depolymerization technically challenging and economically inefficient [16]. To overcome these challenges, thermochemical processes (e.g., gasification and pyrolysis) have attracted attention as viable options. These processes can handle mixed and contaminated plastic streams by thermally breaking down polymers into simpler products, including oil, combustible gases (*e.g.*, syngas), and char. Pyrolysis, conducted at intermediate temperatures (450–800 ℃) under an inert atmosphere, primarily yields pyrogenic oil [17]. Although this oil has potential as an energy source and chemical feedstock, its complex composition creates challenges during direct utilization without further upgrading steps, such as separation or catalytic reforming [18]. These downstream processing requirements can adversely affect the overall economic viability of pyrogenic oil application [19]. Meanwhile, gasification operates at high temperatures (800–1000 ℃) in the presence of oxidizing agents (*e.g.*, air, oxygen, and steam) [20] and mainly produces syngas, a mixture of H_2 and CO, that can be used for power production or as a feedstock in chemical synthesis [21]. Compared to pyrogenic oil, syngas can be easily separated and purified [22]. However, gasification is energy-intensive and requires high temperatures to proceed.

In this context, this study aimed to maximize syngas recovery by promoting the conversion of condensable intermediates into non-condensable gases at pyrolysis temperature (≤700 ℃), offering a potentially sustainable alternative to conventional high-temperature gasification. PBT was selected as a model plastic representing ELV-derived materials, particularly from the automobile sector. PBT is extensively used in interior and exterior vehicle components, including fuel systems, connectors, and gears, owing to its superior thermal and mechanical properties [23]. Although numerous studies have investigated the pyrolysis of various plastics in ELVs to produce gaseous and liquid products [24,25], to the best of our knowledge, PBT waste has not been specifically investigated for syngas-oriented pyrolysis. To enhance both the environmental and technical benefits of the process, CO_2 was employed as a reaction medium during pyrolysis, and its role was elucidated by comparing product distributions under inert and CO_2 atmospheres. Unlike combustion, which generates substantial CO_2 emissions and potential toxic byproducts, using CO_2 as a reactant provides route for carbon utilization and may mitigate environmental burdens. Furthermore, the introduction of a catalyst allows for efficient chemical cracking at moderate temperatures, significantly reducing energy consumption compared to the extreme heat required for traditional gasification. The core innovation of this work lies in the integrating a nickel-based catalyst to facilitate CO_2 activation during pyrolysis. The nickel-based catalyst was selected because of its economic advantage over noble metals and its proven ability to break C–C, C–H, and C–O bonds [26]. For practical application in industrial-scale reactors, catalytically active components are typically supported on porous pellets to minimize internal pressure drop. Accordingly, an eggshell-type catalyst was employed. By facilitating CO_2 activation, this system maximizes syngas recovery while maintaining moderate process temperatures, effectively addressing the limitations of both traditional gasification and non-catalytic pyrolysis. To further clarify the mechanistic role of CO_2 under catalytic conditions, terephthalic acid (TPA), a representative PBT monomer, was employed as a model compound. Overall, this study represents a significant advancement by demonstrating a synergistic CO_2-catalyst system that enables a dramatic enhancement in syngas recovery, far exceeding conventional pyrolysis performance. Furthermore, it provides a pioneering mechanistic framework to explain CO_2 activation, offering a new perspective on sustainable plastic waste-to-energy conversion.

2. Materials and methods

2.1. Sample and chemical reagents

PBT waste was obtained from a plastic recycling facility located in Seongdong-gu, Seoul, Korea. The facility processes end-of-life plastic streams collected from industrial sources; the PBT fraction was automatically sorted, washed, and shredded prior to sampling. Benzoic acid (C_6H_5COOH, ≥99.5%), TPA (C_6H_4-1,4-$[CO_2H]_2$, 98.0%), and PAH standards (EPA 525 PAH Mix A) were obtained from Sigma-Aldrich (USA). Dichloromethane (DCM; CH_2Cl_2, 99.8%), activated alumina (ɣ-Al_2O_3, ≤3-mm diameter), nickel nitrate hexahydrate ($Ni[NO_3]_2 \cdot 6\ H_2O$, 97%), and 1-octanol ($C_8H_{17}OH$, ≥98%) were purchased from DAEJUNG Chemical (Korea). Ultra-high-purity N_2, air, and CO_2 (99.999%) were acquired from Green Gas (Korea). Additionally, 20-vol% H_2 reducing gas (balanced with N_2) and a standard gas mixture comprising 20-vol% H_2, CO, CH_4, CO_2, and N_2, were purchased from RIGAS (Korea).

2.2. Characterization of PBT

The specific functional groups in PBT samples were determined using Fourier transform infrared spectroscopy (FT-IR; Cary 630, Agilent Technologies, USA) together with an attenuated-total-reflectance module. The thermal degradation behavior of PBT was examined using thermogravimetric analysis (TGA; STA 449 F5 Jupiter, Netzsch, Germany), and TGA tests were conducted in the temperature range of 40–900 ℃ and at a heating rate of 10 ℃ min^{-1}, using 10 ± 0.1 mg of PBT per run. The total gas flow rate was maintained at 100 mL min^{-1}, comprising 80 mL min^{-1} of purge gas (N_2 and CO_2) and 20 mL min^{-1} of protective gas (N_2) inside a sealed reaction chamber. Before the TGA tests, a blank run was conducted to exclude the influence of mass changes due to buoyancy effects at high temperatures.

2.3. Preparation and characterization of Ni-based catalyst

An eggshell-type Ni catalyst was synthesized by depositing a thin layer of Ni species onto the outer surface of ɣ-Al_2O_3, following a previously reported method [27]. First, ɣ-Al_2O_3 was dried at 450 ℃ for 2 h under N_2 to remove moisture. To promote preferential deposition on the external surface, the dried support was then immersed in 1-octanol for 24 h, rendering the internal pores hydrophobic. Thereafter, excess 1-octanol was removed via filtration. A Ni precursor solution was prepared by dissolving $Ni(NO_3)_2 \cdot 6\ H_2O$ in deionized water to obtain 5-wt% Ni. This solution was added dropwise onto the treated ɣ-Al_2O_3 at 130 ℃, allowing the precursor to become immobilized as the water evaporated. Subsequently, the catalyst was subjected to thermal treatment. First, the hydrophobic reagent was removed by heating at 200 ℃ for 2 h, considering the boiling point of 1-octanol (196 ℃). Calcination was then performed under an air atmosphere at 750 ℃ for 3 h. Finally, the calcined catalyst was reduced under H_2 (20-vol% H_2 balanced with N_2) at 750 ℃ for 3 h using a heating ramp of 2 ℃ min^{-1}. The final Ni content of the catalyst was 4.93 wt%, determined using inductively coupled plasma optical emission spectrometer (5800, Agilent, USA).

After catalytic pyrolysis under both N_2 and CO_2 conditions, the spent catalysts were recovered. Temperature-programed oxidation (TPO) analyses were conducted on the fresh catalyst and the two spent catalysts using a TGA instrument. Approximately 30 ± 0.1 mg of each catalyst was placed on the sample holder, and the reactor was purged with a gas mixture consisting of 80 mL min^{-1} of air and 20 mL min^{-1} of N_2. The TGA test proceeded in two sequential steps: (1) impurity removal, carried out by heating from 40 to 200 ℃ at a rate of 20 ℃ min^{-1} followed by a 30-minute hold at 200 ℃, and (2) the TPO measurement, performed by heating from 200 to 900 ℃ at the same ramp rate. During the TPO step, the oxidation of carbon residues on the used catalysts resulted in a decrease in sample mass.

To evaluate the stability of the catalyst, comparative characterizations were performed on both the fresh and spent catalysts (after CO_2-cofeeding pyrolysis). The reducibility profiles were obtained by H_2-temperature programmed reduction (H_2-TPR) using a BELCAT II instrument (MicrotracBE, Japan). The surface morphology and elemental distribution were investigated using scanning electron microscopy coupled with energy dispersive spectroscopy (SEM-EDS) on a CUBE II instrument (EMCraft, Korea). The crystal structures were elucidated through X-ray diffraction (XRD) using a D/Max-2500 instrument (Rigaku, Japan). All experiments were conducted in triplicate, and the relative standard deviation (RSD) of the key response variables was below 2%. Accordingly, error bars were omitted for clarity.

2.4. Experimental setups for pyrolysis

The pyrolysis experiments were conducted using PBT and its moder compound, TPA. All experiments were performed in a tubular quartz reactor (inner diameter: 22 mm; outer diameter: 25 mm; length: 1000 mm) contained within a cylindrical furnace equipped with two independently controlled heating zones. All experiments were conducted at atmospheric pressure. Prior to each run, the reactor was purged with feed gas (N_2 or CO_2) for 30 min to remove residual air. The set temperatures were monitored using K-type thermocouples positioned at the center of the sample and catalyst bed. To ensure a stable temperature in the downstream heating zone, the first-stage furnace was preheated to 100 ℃ and held for 30 min prior to each experiment. Subsequently, the second-stage furnace was adjusted to 700 ℃ and held for 30 min before initiating the programmed heating of the primary zone.

The following three pyrolysis configurations were used for PBT: one-stage, two-stage, and catalytic pyrolysis (Fig. 1). For each experiment, 2.00 ± 0.01 g of PBT was loaded at the center of the first heating zone under constant N_2 (800 mL min^{-1}). In the one-stage pyrolysis setup, the furnace was heated from 100 to 700 ℃ at a heating rate of 10 ℃ min^{-1}. For two-stage pyrolysis, an auxiliary heating zone was added downstream of the primary zone, operating isothermally at 700 ℃ to provide additional thermal energy and promote further cracking of volatile intermediates. In the catalytic pyrolysis configuration, the same temperature profile as that used for two-stage pyrolysis was maintained. Additionally, 4.00 ± 0.05 g of a Ni-based catalyst was placed at the center of the second heating zone. All the PBT pyrolysis experiments were conducted using 80 vol% of CO_2 (balanced with N_2), with the total gas flow rate maintained at 800 mL min^{-1}. For TPA, catalytic pyrolysis was conducted exclusively, using 1.00 ± 0.01 g of TPA and 2.00 ± 0.01 g of the Ni-based catalyst under both N_2 and 80-vol% CO_2 conditions, with the same temperature profile as those in the catalytic PBT experiments. These conditions were selected to assess the CO_2-induced thermal degradation mechanistic pathways in the presence of the catalyst.

2.5. Characterization of pyrogenic products

PBT pyrolysis generates volatiles consisting of condensable and non-condensable pyrogenic products. Volatiles exiting the quartz reactor were first passed through a cold trap, where the condensable fraction, referred to as pyrogenic oil, was collected using a condensation system maintained at −40 ℃. Thereafter, the collected condensable products were dissolved in 100-mL DCM. The resulting solution was analyzed using a gas chromatography (GC)/mass spectrometry (GC/MS, 8890/5977 B, Agilent, USA) unit and GC/flame ionization detection (GC/FID, 8890, Agilent, USA), and both units were equipped with a DB-5MS column. Quantitative analysis of the identified condensable products was performed via calibration with standard solutions of benzoic acid, TPA, and a PAH mixture. The concentrations of the compounds not included in the standard solution were estimated by applying a relative response factor based on the effective-carbon-number method [28]. Following condensation, the non-condensable gases, including H_2, CO, and $C_{1–2}$ HCs, were analyzed using a micro-gas chromatograph (micro-GC, 990, Agilent, USA). Calibration curves used for quantitative analysis are provided in the Support data (Fig. S1–2). The micro-GC was equipped with two columns: a Plot U capillary column (8 m ×320 μm ×30 μm) and a Molsieve 5 A packed column (10 m ×320 μm ×30 μm). This analytical procedure was applied to both the PBT and TPA catalytic pyrolysis experiments to evaluate the effect of CO_2 conditions on the distribution of volatile products. Detailed operating conditions for

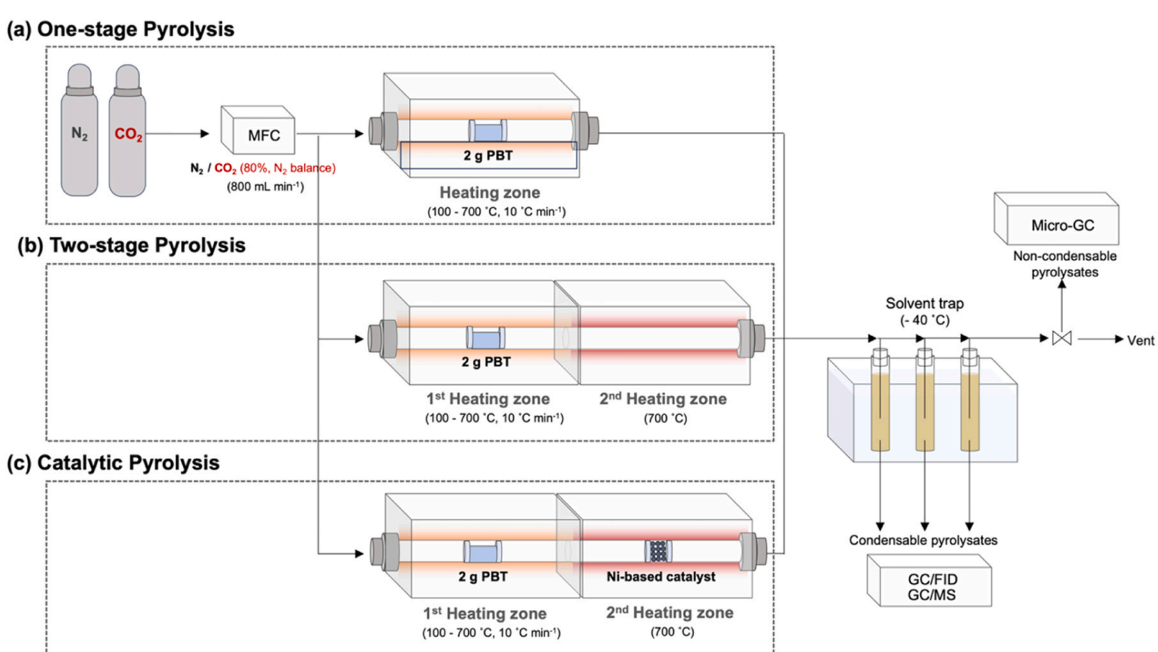

Fig. 1. Schematic diagram of the experimental setups.

GC/MS, GC/FID, and micro-GC analyses are provided in the Table S1. All experiments were performed in triplicate (independent runs). The RSD of the key response variables was < 2%, and thus, error bars were omitted for clarity, and the corresponding standard deviations are provided in the Support data (Fig. S3-7)

3. Results & discussions

3.1. Characterization of PBT

Before the pyrolysis experiments, the PBT samples were characterized using FT-IR and TGA analysis. FT-IR analysis was performed to identify the characteristic functional groups present in the polymer. The FT-IR spectrum of PBT exhibited several distinct peaks: a peak at 2955 cm^{-1}, corresponding to the C–H stretching vibrations of both saturated and unsaturated groups; a strong absorption peak near 1707 cm^{-1}, attributed to the stretching vibration of the ester carbonyl group (C=O); a peak at 1451 cm^{-1}, due to C–H bending in the methylene (CH_2) group; and a peak at 1408 cm^{-1}, associated with the C=C stretching vibration in the aromatic ring structure (Fig. 2(a)). Sharp peaks at 1244 and 1093 cm^{-1} were assigned to C–O stretching in ester groups and O–CH_2 stretching, respectively. In addition, the peak at 1017 cm^{-1} was related to in-plane aromatic C–H bending, while those at 877 and 726 cm^{-1} corresponded to out-of-plane C-H bending and out-of-plane vibrations of the terephthalate unit, respectively [29,30].

To assess the thermolytic behavior of PBT, TGA was performed under inert (N_2) conditions. Testing was conducted over a temperature range of 40–900 ℃ and at a heating rate of 10 ℃ min^{-1}. Fig. 2(b) shows the mass loss profile and corresponding differential thermogram (DTG) curves. The observed mass loss was primarily attributed to the release of volatile organic compounds (VOCs) during the thermal decomposition of PBT. The sharp weight reduction that occurred between 340 and 440 ℃ suggested the emission of thermally decomposed VOCs. The DTG curve exhibited a single peak at 403 ℃, indicating that the decomposition of PBT proceeded mainly via a single-stage decomposition reaction. At ≥ 440 ℃, the decomposition rate decreased. The final mass of the residue obtained at 900 C was 7.27 wt%. This was likely due to the dehydrogenation of carbon-based materials, which leads to the formation of solid residues through carbonization [31]. Although carbonization phenomena are typically associated with biomass pyrolysis, similar processes are expected in PBT, especially given its aromatic backbone, as shown in Fig. 2(a) [32]. Importantly, the degree of aromaticity in this material is known to increase with temperature, as dehydrogenation reactions are more extensive under higher-temperature conditions [33]. Therefore, it is reasonable to infer that the similar structural characteristics of PBT facilitated the formation of solid carbon residues during thermal decomposition.

3.2. Conventional pyrolysis of PBT

Building upon the thermal decomposition characteristics observed in the TGA test, one-stage pyrolysis was conducted to assess the evolution of VOCs under practical reaction conditions. The major condensable products were oxygenated aromatics, primarily benzoic acid derivatives (*e.g.*, 4-ethylbenzoic acid) and phthalic acid derivatives (*e.g.*, dibut-3-enyl terephthalate and TPA) (Fig. 3(a)). These compounds reflected the stepwise degradation mechanism of PBT, as illustrated in the reaction pathway shown in Fig. 3(b). Cyclization occurs at 270–280 ℃ via intramolecular exchange process, leading to the formation of cyclic oligomers [34,35]. With increases in the temperature to approximately 290 ℃, these cyclic structures undergo ring-opening reactions triggered by β-CH hydrogen transfer, producing unsaturated oligomers (*e.g.*, dibut-3-enyl terephthalate and 4-(benzoyloxy)butyl but-3-enyl terephthalate) [34,35]. At temperatures exceeding 300 ℃, these unsaturated intermediates subsequently undergo secondary fragmentation, including dehydrogenation, decarboxylation, and side-chain cleavage [36]. As a result, these reactions produce low-molecular-weight aromatic acids, primarily benzoic acid derivatives with functional substitutions (*e.g.*, methyl, ethyl, and benzoyloxy groups), as well as phthalic acid derivatives formed through the partial cleavage of the ester linkages in PBT. Concurrently, gaseous products, such as H_2, CO, CO_2, and light HCs ($C_{1–2}$ alkanes and alkenes), are released, most likely as a result of the aforementioned secondary fragmentation reactions. Detailed concentration profiles for the evolved gaseous products are shown in Fig. 4.

Except H_2, all gaseous products (CO, $C_{1–2}$HCs, and CO_2) began evolving at temperatures above 370℃. Notably, the evolution range of these gases closely matched the mass loss observed in the TGA profile (Fig. 3(b)), indicating that the gas evolution coincided with the thermal decomposition of the polymeric backbone at 340–440 ℃. CO formation was attributed to the radical decarbonylation of carbonyl-containing intermediates [37]. Meanwhile, CO_2 was likely produced via the concerted decarboxylation of aromatic acid intermediates such as TPA [37]. $C_{1–2}$HCs likely originated from the thermal cleavage of aliphatic side chains (*e.g.*, the butylene segment [-$CH_2CH_2CH_2CH_2$-]) [38]. On the other hand, H_2 evolution began at approximately 430 ℃ and continued to progress steadily until a temperature of 700 ℃ was reached. This trend corresponds with the TGA results in Fig. 3(b), suggesting that hydrogen

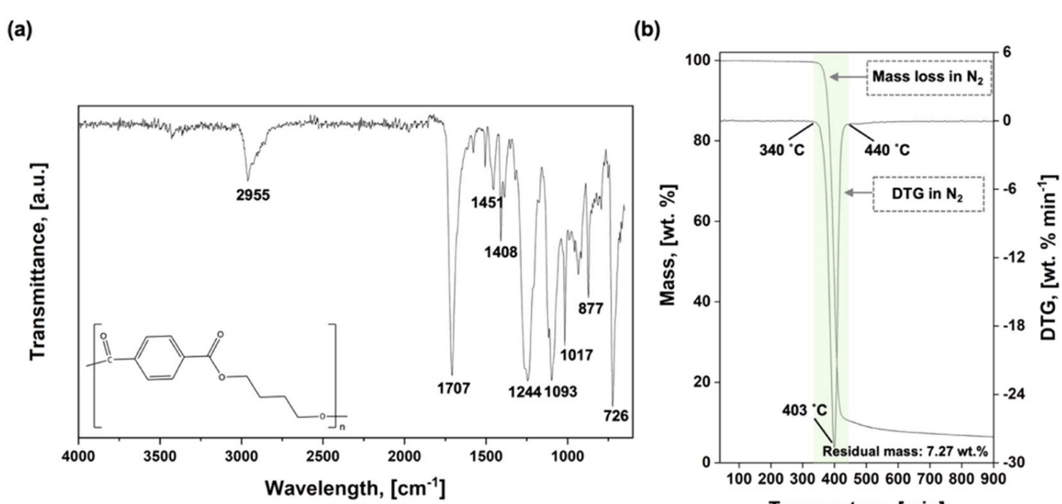

Fig. 2. (a) FT-IR spectrum and (b) mass change profiles, together with the corresponding differential thermograms (DTGs), for PBT.

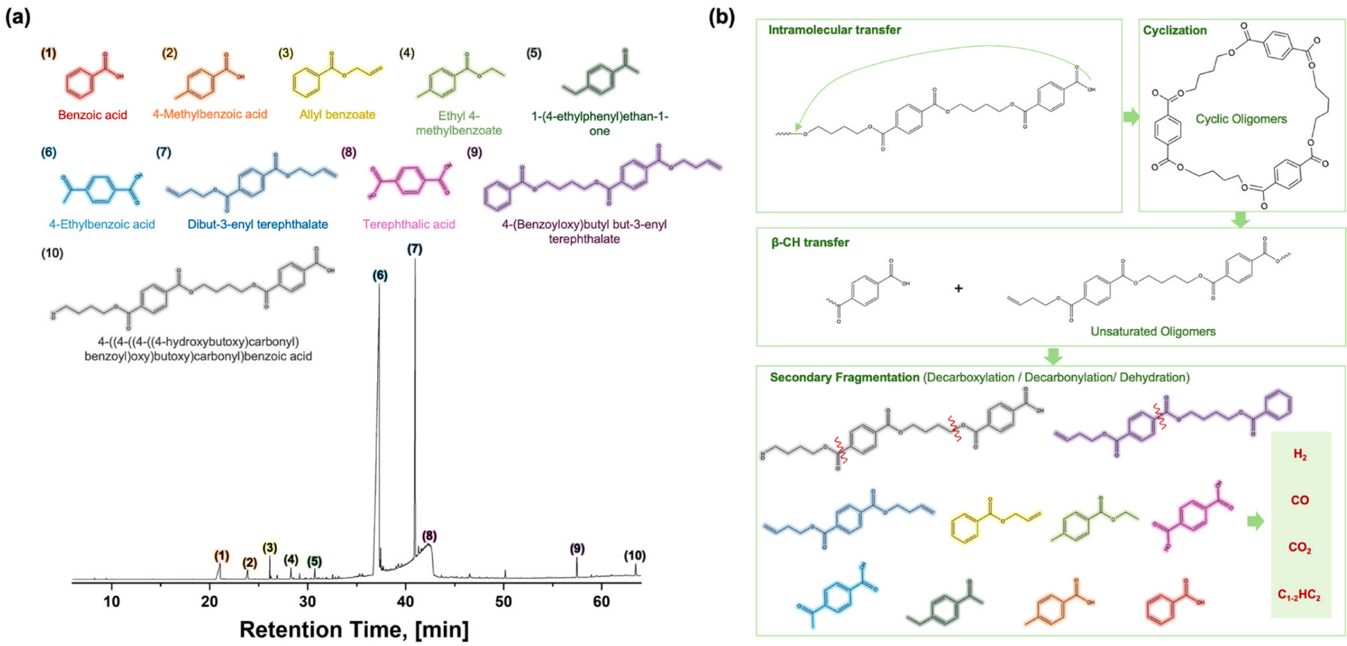

Fig. 3. (a) Chromatogram of condensable products resulting from the one-stage pyrolysis of PBT under inert conditions, and (b) proposed reaction pathway.

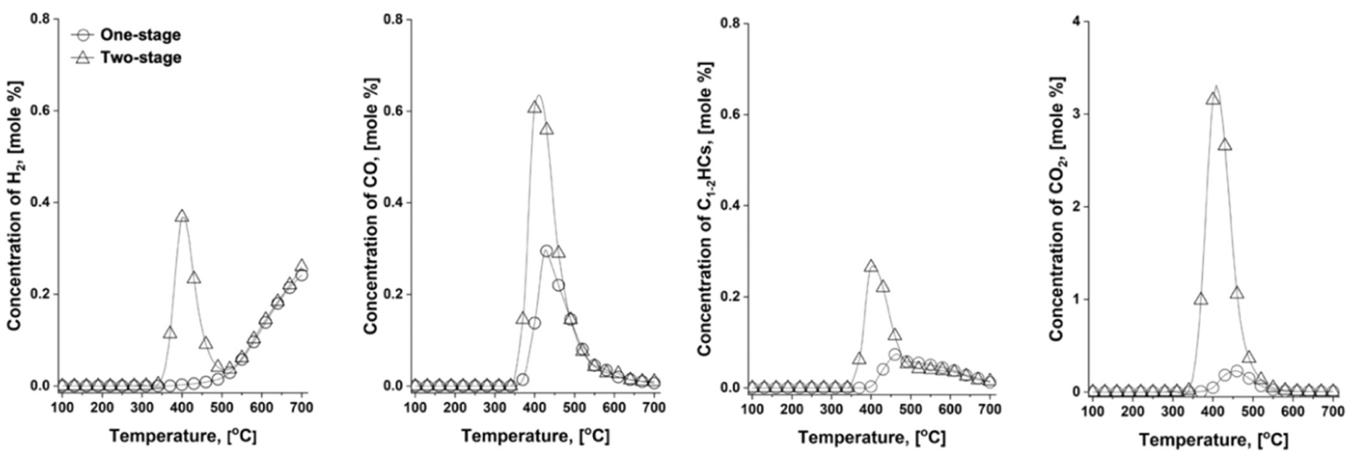

Fig. 4. Concentration profiles of major gaseous products (H_2, CO, C_{1-2}HCs, and CO_2) that evolved from one-stage (circles) and two-stage (triangles) pyrolysis under inert conditions.

release was driven by progressive dehydrogenation, which subsequently contributed to the carbonization of the polymer matrix at elevated temperatures [33]. Interestingly, the concentration of C_{1-2}HCs began decreasing as the H_2 concentration sharply increased above 520 ℃. This inverse relationship suggested that C_{1-2}HCs may have served as hydrogen donors via dehydrogenation [39].

Nevertheless, the overall yield of gaseous products remained negligible, indicating that most of the carbon in PBT was converted into condensable rather than gaseous products. Excluding char formation (7.83 wt% of PBT), most of the remaining carbon was ascribed to condensable products, which accounted for 67.91 wt%. Notably, a substantial portion of these condensable products consisted of oxygenated aromatics such as benzoic acid derivatives and phthalic acid derivatives (Fig. 3(a)). These compounds are well known to cause operational challenges such as reactor clogging, corrosion, and reduced fuel quality in downstream processing [40]. Therefore, to improve the practical applicability of PBT waste pyrolysis, it is crucial to redirect carbon fluxes toward gaseous products (especially syngas), rather than condensable products, which can hinder process efficiency.

Evidently, supplying additional thermal energy facilitates the further conversion of condensable products into gaseous products. To implement this process as a strategy, the one-stage pyrolysis system was reconfigured into a two-stage setup. An auxiliary heating unit, operated at a constant temperature of 700 ℃, was installed downstream of the primary pyrolysis reactor to provide additional heat. Temperature substantially lower than this are generally insufficient for effective vapor upgrading, whereas excessively high temperatures can increase energy demand and may intensify secondary char/coke formation [41,42]. Thus, 700 ℃ was selected as a practical compromise and was held constant to ensure comparability across configurations. As shown in Fig. 4, the concentration profiles of the major gaseous products generated from the two-stage pyrolysis of PBT were compared with those from the one-stage configuration. The results indicated that the concentrations of H_2, CO, C_{1-2}HCs, and CO_2 increased by approximately 2-, 2-, 2-, and 12-fold, respectively, under two-stage pyrolysis.

Along with the enhanced formation of gaseous products, significant changes were observed in the composition of the condensable products. The chromatograms of the condensable products obtained from one-stage and two-stage pyrolysis are compared in Fig. 5(a). The total masses of the major chemical compounds are shown in Fig. 5(b). Fig. 5(b) also shows the yields of the chemical classes based on mass. These compounds were categorized as benzoic acid derivatives, phthalic acid derivatives, PAHs, simple aromatics, and others. Detailed information regarding the identified chemical compounds is provided in Table 1. One of the most notable findings obtained from the two-stage pyrolysis was the substantial reduction in the overall yield of condensable products (from 67.91 to 35.39 wt%). Specifically, during two-stage pyrolysis, the relative abundance of high-molecular-weight compounds (*e.g.*, 4-ethylbenzoic acid, dibut-3-enyl terephthalate, TPA, and 4-(benzoyloxy)butyl but-3-enyl terephthalate) decreased. In contrast, the relative abundance of lower-molecular-weight compounds (*e.g.*, toluene, styrene, benzoic acid, and 4-methylbenzoic acid) and PAHs (naphthalene, 2-methylnaphthalene, 1-methylnaphthalene, acenaphthylene) increased (Fig. 5(b) and Table 1). This shift was primarily attributed to the enhanced thermal severity in the second-stage reactor, which promoted the further cracking of unconverted intermediates from the first-stage pyrolysis. Under these conditions, secondary reactions such as dehydrogenation, decarbonylation, decarboxylation, hydrogen abstraction/acetylene addition, and Diels–Alder cycloaddition (leading to crosslinking) were thermally facilitated [43,44]. These reactions transformed high-molecular-weight compounds into lower-molecular-weight compounds and PAHs, while also contributing to the evolution of gaseous products (H_2, CO, CO_2, and C_{1-2}HCs) (Fig. 4). In particular, the 12-fold increase in CO_2 concentration during two-stage pyrolysis strongly suggested that decarboxylation was the predominant secondary reaction pathway. This was corroborated by the notable reduction in the contents of carboxylic-acid-bearing intermediates. The concentrations of 4-ethylbenzoic acid and TPA decreased from 24.83 to 0 wt% and from 32.02 to 22.46 wt%, respectively (Table 1).

Although thermal energy has been identified as a key factor promoting the conversion of condensable products into gaseous products, a considerable fraction (35.39 wt%) of condensable products remained, primarily composed of phthalic acid derivatives and benzoic acid derivatives. This persistence may be attributed to the inherent aromatic stability of these compounds, which inhibits their thermal degradation [45]. To overcome this limitation, the use of nickel-based catalysts (Ni/Al_2O_3) has been proposed as a promising strategy to enhance syngas production from PBT waste. Nickel-based catalysts are widely recognized as cost-effective solutions for tar reforming, owing to their strong ability to activate the cleavage of C–C and C–H bonds in aromatic structures, thereby promoting their conversion into syngas [46,47]. To accurately assess the catalytic effect, the concentration profiles of the gaseous products that evolved from catalytic pyrolysis over a Ni-based catalyst were compared with those obtained from non-catalytic

Table 1
Major constituents of pyrogenic oil obtained from PBT pyrolysis (one-stage and two-stage pyrolysis) under inert conditions.

No.	Retention time (min)	Chemical compounds	Formula	Yield (wt%)	
				One-stage	Two-stage
1	7.16	Toluene	C_7H_8	N.D.	0.08
2	11.67	Styrene	C_8H_8	N.D.	0.10
3	17.08	1-Ethynyl-4-methylbenzene	C_9H_8	N.D.	0.09
4	21.47	Naphthalene	$C_{10}H_8$	N.D.	0.30
5	21.76	Benzoic acid	$C_7H_6O_2$	0.93	9.14
6	23.88	4-Methylbenzoic acid	$C_8H_8O_2$	0.29	1.12
7	24.61	2-Methylnaphthalene	$C_{11}H_{10}$	N.D.	0.07
8	25.08	1-Methylnaphthalene	$C_{11}H_{10}$	N.D.	0.06
9	26.13	Allyl benzoate	$C_{10}H_{10}O_2$	0.19	N.D.
10	26.32	4-Ethylbenzoic acid	$C_9H_{10}O_2$	0.05	0.11
11	26.89	2-Vinylnaphthalene	$C_{12}H_{10}$	0.05	N.D.
12	27.09	4-Vinylbenzoic acid	$C_9H_8O_2$	N.D.	0.59
13	28.29	Ethyl 4-methylbenzoate	$C_{10}H_{12}O_2$	0.09	n.d.
14	28.66	Acenaphthylene	$C_{12}H_8$	N.D.	0.06
15	29.17	4-(*tert*-Butyl)phenol	$C_{10}H_{14}O$	0.03	N.D.
16	30.73	4'-Ethylacetophenone	$C_{10}H_{12}O$	0.10	N.D.
17	32.55	3-Ethoxyphthalide	$C_{10}H_{10}O_3$	0.05	N.D.
18	34.00	3,4-Dihydronaphthalene-2-carboxylic acid	$C_{11}H_{10}O_2$	N.D.	0.17
19	34.33	2-Naphthoic acid	$C_{11}H_8O_2$	N.D.	0.91
20	37.25	4-Acetylbenzoic acid	$C_9H_8O_3$	24.83	N.D.
21	38.30	Biphenyl-4-carboxylic acid	$C_{13}H_{10}O_2$	N.D.	0.13
22	40.97	Dibut-3-enyl terephthalate	$C_{16}H_{18}O_4$	8.38	N.D.
23	41.52	Terephthalic acid	$C_8H_6O_4$	32.02	22.46
24	50.15	Di(but-3-en-1-yl) phthalate	$C_{16}H_{18}O_4$	0.13	N.D.
25	57.47	4-(Benzoyloxy)butyl but-3-enyl terephthalate	$C_{23}H_{24}O_6$	0.39	N.D.
26	63.47	4-((4-((4-((4-hydroxybutoxy)carbonyl)benzoyl)oxy)butoxy)carbonyl)benzoic acid	$C_{24}H_{24}O_8$	0.38	N.D.
Pyrogenic oil yield (wt%)				67.91	35.39

*N.D.: Not Detected.
*The data presented corresponds to a representative run selected from triplicate experiments. The major compounds showed high reproducibility with a relative standard deviation (RSD) of < 5%.

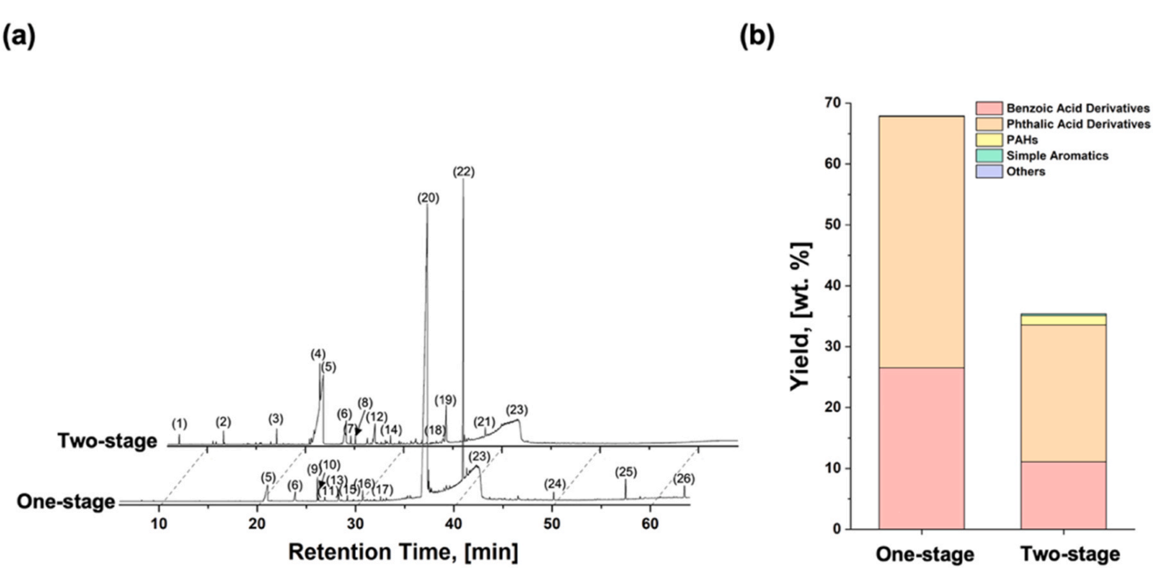

Fig. 5. (a) Chromatograms of condensable products obtained from one-stage and two-stage pyrolysis under inert conditions, and (b) the yield distribution of chemical classes based on mass.

two-stage pyrolysis (Fig. 6).

The most significant finding obtained from catalytic pyrolysis was the strong promotion of dehydrogenation and decarbonylation reactions by the Ni-based catalyst. This was evidenced by the substantial increases in the amounts of H_2 and CO formed (approximately 11- and 7-fold, respectively), compared to those obtained from the two-stage pyrolysis of PBT. In addition to this dominant pathway shift, additional catalytic effects were observed. In the presence of the catalyst, C_{1-2}HC formation was enhanced, which could be attributed to the cleavage of alkyl side chains from the aromatic intermediates. This was facilitated by the ability of the catalyst to activate the C–C and C–H bonds in alkyl-substituted aromatic compounds. In addition, a reduction in CO_2 concentration under catalytic conditions was observed, likely due to two factors: (1) the Ni-based catalyst induced a shift in the decomposition of oxygenated intermediates, favoring decarbonylation over decarboxylation, and (2) it also facilitated the reverse water–gas shift (RWGS) reaction ($H_2 + CO_2 \leftrightharpoons CO + H_2O$) [48]. Therefore, the experimental results clearly demonstrated that the Ni-based catalyst provided an effective means of enhancing syngas production from PBT.

3.3. CO_2-assisted pyrolysis of PBT

Although CO_2 production was reduced under catalytic conditions, CO_2 remained one of the predominant gaseous products that evolved during PBT pyrolysis. Given that CO_2 is a potential greenhouse gas, reducing its emissions is a critical global challenge, especially in sectors such as the automobile industry, which faces increasing pressure to reduce its carbon footprint across both production and end-of-life stages [49,50]. In this study, however, CO_2 was not merely considered an inevitable byproduct but was intentionally introduced as a reactive gas medium, and its mechanistic involvement in the pyrolysis process was experimentally investigated. This approach is consistent with recent advances in plastic waste gasification, in which CO_2 is utilized to promote syngas formation and pollutant control [51]. Unlike gasification, which requires high temperatures (700–1000 ℃) [52], this study investigated the mechanistic role of CO_2 under moderate pyrolysis conditions (below 700℃), aiming to understand how it can facilitate carbon transformation pathway processes. Such an approach not only reduces net CO_2 emissions but also enhances the circular value of waste plastics and CO_2 itself via integrated resource recovery. To this end, PBT was pyrolyzed under CO_2 conditions.

To investigate the thermolytic behavior of PBT under CO_2 conditions, TGA was performed (Fig. 7(a)). Below 790 ℃, the thermal decomposition profiles of PBT under the two conditions were identical. This suggested that no heterogeneous reaction between the solid PBT and CO_2 occurred within this temperature range. Heterogeneous reactions generally appear as distinct shifts in the thermal degradation patterns during TGA, which were not observed here. However, above 790℃, a marked difference emerged. A significant additional mass change occurred under CO_2 conditions, resulting in final residual masses of 7.27 wt% and 3.88 wt% under N_2 and CO_2 conditions, respectively. This disparity provided strong evidence of a heterogeneous reaction between PBT and CO_2, most likely due to the Boudouard reaction ($C + CO_2 \leftrightharpoons 2CO$), a gasification process that becomes thermodynamically favorable at temperatures exceeding 710 ℃ [53]. Therefore, it was concluded that no heterogenous reaction occurred below 790 ℃.

To further explore the role of CO_2 in homogenous gas-phase reactions, the non-catalytic pyrolysis of PBT was conducted under two different configurations (one-stage and two-stage) in the presence of CO_2. Interestingly, while CO formation consistently increased at temperatures above 550 ℃ in both setups (Fig. 6(b)), no significant changes were observed in the concentration profiles of other gaseous products (such as H_2, C_{1-2}HCs, and CO_2). Additionally, the yield and composition of condensable products remained nearly unchanged relative to those under N_2 conditions. The data for the unchanged species were omitted for clarity. These results indicated that CO_2 indeed participated in the gas-phase reaction; however, its influence was limited to enhancing CO formation rather than altering heavier compounds. To interpret this behavior, we compared it with previous findings for the polyoxymethylene copolymer (POM-c) [54]. Under high-temperature and high-density conditions, CO_2 deviates from its linear geometry and adopts a transient dipole moment; this increases its reactivity toward radical or electrophilic reactions [55]. In the case of POM-c, the unstable hydroxyl end groups facilitate spontaneous unzipping depolymerization via a radical mechanism, which is triggered at relatively low temperatures compared to those of other plastics [56]. The resulting alkyl and alkoxy radicals are presumed to react with CO_2, potentially undergoing deoxygenation to release CO and/or CO_2 as gaseous products [57]. Contrastingly, PBT possesses a more thermally stable and rigid aromatic backbone and, thus, requires a higher activation energy for radical fragmentation [36]. Consequently, fewer reactive radicals are formed during pyrolysis, which limits the extent to which CO_2 can participate in secondary gas-phase reactions. This structural difference likely explains why, despite the observed increase in CO formation, the composition of condensable products remained largely unaffected under CO_2 conditions.

Given that CO_2 had a limited influence on the condensable products under non-catalytic conditions, catalytic pyrolysis using a Ni-based

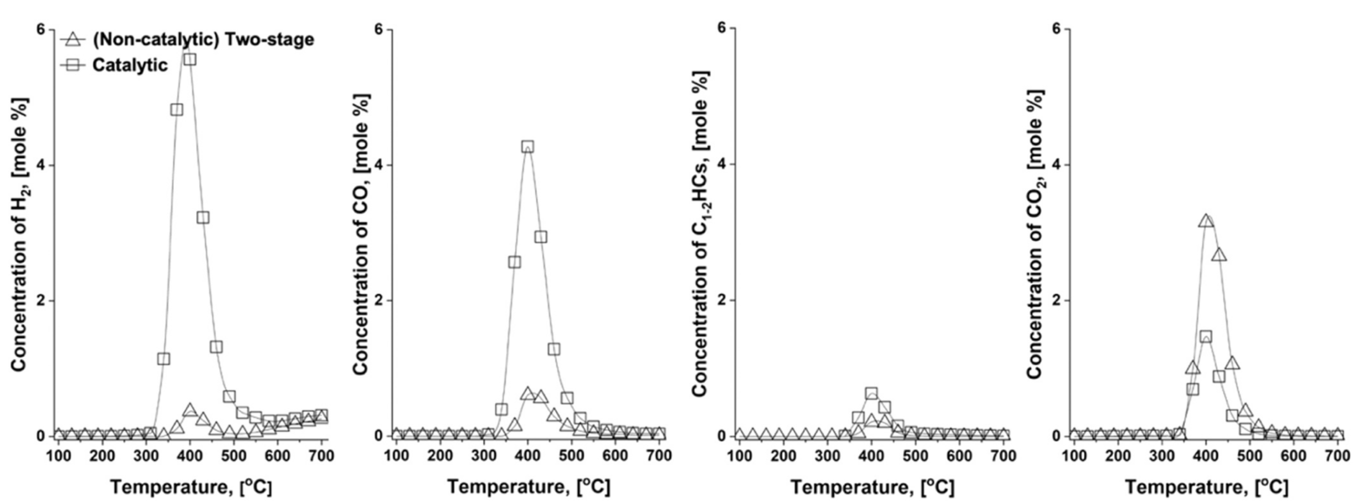

Fig. 6. Concentration profiles of major gaseous products (H_2, CO, C_{1-2}HCs, and CO_2) that evolved from two-stage (triangles) and catalytic (squares) pyrolysis under inert conditions.

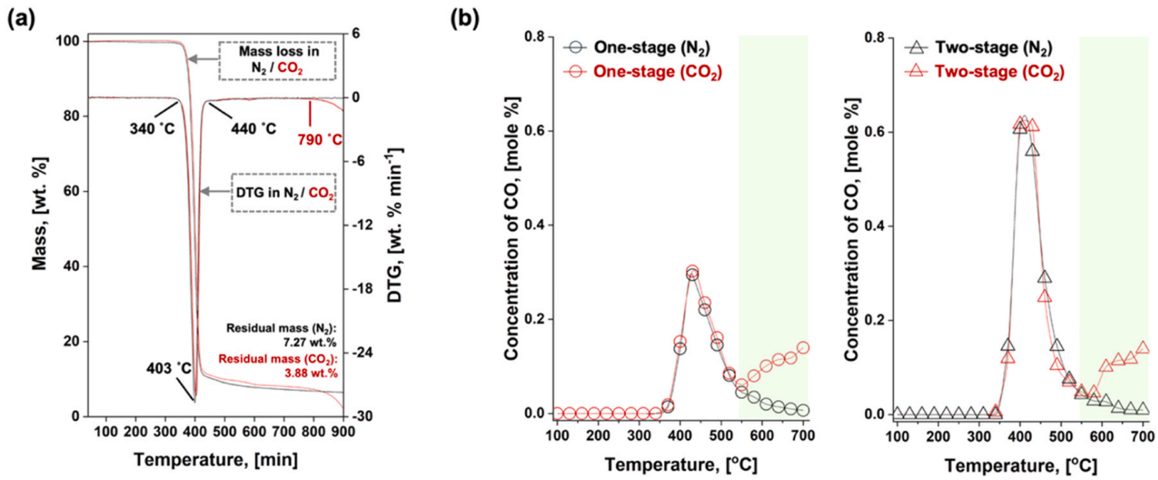

Fig. 7. (a) Mass change profile and the corresponding differential thermograms (DTGs) for PBT under N_2 (black line) and CO_2 (red line) conditions, and (b) concentration profiles of CO evolved from one-stage (circles) and two-stage (triangles) pyrolysis under N_2 (black) and CO_2 (red) conditions.

catalyst was employed to enhance its reactivity. Noble-metal catalysts (*e.g.*, Pt, Pd, and Rh) are frequently reported to deliver high activity and improved resistance to deactivation in HC reforming and oxygenate conversion. However, their high cost and scarcity can be prohibitive for large-scale applications. In this context, Ni is substantially less expensive and more abundant, making Ni-based catalysts economically attractive for scalable processes while still providing competitive activity. Therefore, Ni was selected in this work primarily for cost-effectiveness and practical scalability. The concentration profiles of the gaseous products that evolved from the catalytic pyrolysis under N_2 and CO_2 conditions are compared in Fig. 8. Among the major gas components, only CO and CO_2 differed significantly under CO_2 conditions, whereas H_2 and C_{1-2}HCs remained nearly unchanged. Consistently, the H_2/CO ratio decreased under CO_2 relative to N_2 for all configurations (Table S2), dropping from 0.85 to 1.29 in N_2 condition to 0.24–0.31 in CO_2 condition. This trend reflects that the pronounced CO enrichment under CO_2-assisted catalytic pyrolysis. Unlike non-catalytic pyrolysis where a CO enhancement was observed only at temperatures ≥ 550 ℃, catalytic pyrolysis led to a sharp (five-fold) increase in CO generation from 340℃. This increase was accompanied by a substantial reduction in the CO_2 concentration over the same temperature range, suggesting active CO_2 consumption. Although radical-driven gas-phase reactions between CO_2 and VOCs derived from PBT may contribute to CO formation, such mechanisms are unlikely to account for the substantial CO_2 consumption observed under catalytic conditions. Notably, a comparable decrease in the CO_2 concentration was not observed under non-catalytic conditions, even though an increase in CO formation occurred. Similarly, while the RWGS reaction may have occurred, the negligible change in H_2 concentration indicated that it was not the dominant route. The Boudouard reaction ($C + CO_2 \leftrightarrows 2CO$) was also ruled out, as the TGA results confirmed that it became significant only above 790 ℃ (Fig. 7(b)), well beyond the temperature range applied in this study. A more plausible explanation is that CO_2, upon activation on the Ni catalyst surface, acted as a strong oxidizing agent. This activated CO_2 likely participated in oxidation reactions with VOCs during PBT pyrolysis. These surface-mediated gas-phase reactions may account for the enhanced CO production and the substantial CO_2 consumption observed under catalytic conditions. This behavior suggested that the system utilized CO_2 as a reaction medium and actively converted it into CO during pyrolysis, thus demonstrating its potential for carbon-negative processing.

To further verify this mechanistic interpretation, the condensable products were analyzed (Fig. 9(a–b) and Table 2). Under CO_2 conditions, the total yield of condensable products markedly decreased from 24.55 wt% under N_2 to 10.28 wt% under CO_2, indicating a shift in carbon allocation toward non-condensable products. This shift likely reflected gas-phase reactions induced by CO_2 that divert carbon from heavy oxygenates toward syngas formation. In terms of the chemical class distribution, phthalic acid derivatives showed a sharp decline in

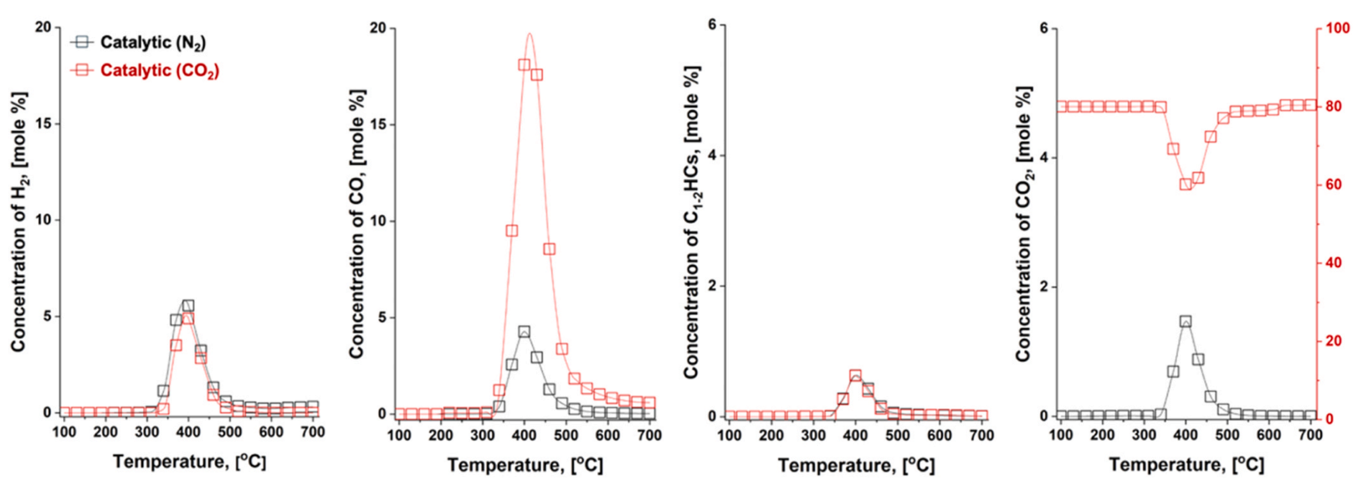

Fig. 8. Concentration profiles of major gaseous products (H_2, CO, C_{1-2}HCs, and CO_2) that evolved from catalytic pyrolysis over a Ni-based catalyst under N_2 (black squares) and CO_2 (red squares) conditions.

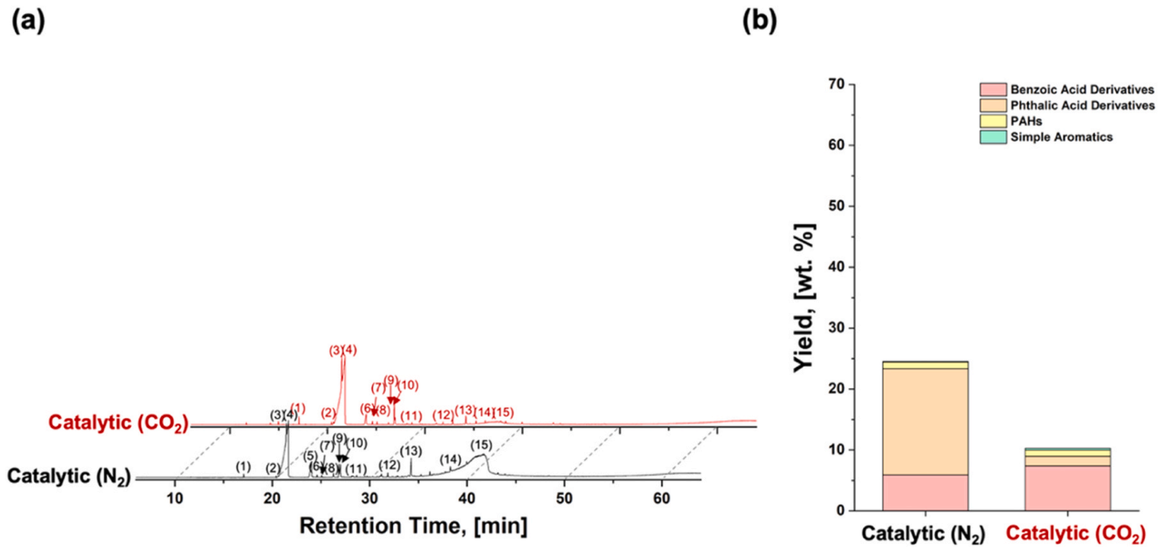

Fig. 9. (a) Chromatograms of condensable products obtained from catalytic pyrolysis under N_2 (black line) and CO_2 (red line) conditions, and (b) the yield distribution of chemical classes based on mass.

Table 2
Major constituents of pyrogenic oil obtained from catalytic PBT pyrolysis under N_2 (black line) and CO_2 (red line) conditions.

No.	Retention time (min)	Chemical compounds	Formula	Yield (wt%)		CO_2/N_2 mass change[a]
				Cata_N_2	Cata_CO_2	
1	17.08	1-Propenylbenzene	C_9H_{10}	0.03	0.06	100.00
2	20.41	2-Methylindene	$C_{10}H_{10}$	0.01	0.02	100.00
3	21.47	Naphthalene	$C_{10}H_8$	0.94	0.91	-3.19
4	21.76	Benzoic acid	$C_7H_6O_2$	4.95	6.91	39.60
5	24.16	4-Methylbenzoic acid	$C_8H_8O_2$	0.60	0.31	-48.33
6	24.61	2-Methylnaphthalene	$C_{11}H_{10}$	0.02	0.03	50.00
7	25.08	1-Methylnaphthalene	$C_{11}H_{10}$	0.02	0.02	0.00
8	26.32	4-Ethylbenzoic acid	$C_9H_{10}O_2$	0.07	0.04	-42.86
9	26.85	Biphenyl	$C_{12}H_{10}$	0.10	0.21	110.00
10	27.09	4-Vinylbenzoic acid	$C_9H_8O_2$	0.27	0.10	-62.96
11	28.66	Acenaphthylene	$C_{12}H_8$	0.02	0.02	0.00
12	31.85	Fluorene	$C_{13}H_{10}$	0.04	0.03	-25.00
13	34.25	2-Naphthoic acid	$C_{11}H_8O_2$	0.35	0.12	-65.71
14	36.18	Phenanthrene	$C_{14}H_{10}$	0.04	0.02	-50.00
15	38.28	Terephthalic acid	$C_8H_6O_4$	17.09	1.47	-91.40
Pyrogenic oil yield (wt%)				24.55	10.28	

*The data presented corresponds to a representative run selected from triplicate experiments. The major compounds showed high reproducibility with a relative standard deviation (RSD) of < 5%.
[a] (Mass in Cata_CO_2 - Mass in Cata_N_2) ÷ Mass in Cata_N_2 × 100

yield under CO_2. For example, TPA, the most abundant phthalic acid derivative under N_2, showed a substantial decrease in yield from 17.09 to 1.47 wt% under CO_2. In contrast, the yield of benzoic acid derivatives increased, with that of benzoic acid rising from 4.95 to 6.91 wt%. Simple aromatic compounds, including prop-1-en-1-ylbenzene, 2-methylindene, and biphenyl, also showed higher yields under CO_2. These compositional changes indicated a clear transformation pathway, whereby high-molecular-weight phthalic acid derivatives were converted into lower-molecular-weight benzoic acid derivatives and light aromatics, which was likely influenced by the oxidative properties of CO_2 in the presence of the Ni-based catalyst. However, because the structural complexity of the condensable products of PBT may mask specific mechanistic aspects, it was difficult to attribute these changes to specific mechanistic features. Therefore, to elucidate more clearly the role of CO_2, TPA was selected as a model compound for further investigation. Owing to its simpler structure and high thermal reactivity, the catalytic pyrolysis of TPA enabled a more precise interpretation of the CO_2-induced reaction pathway.

Fig. 10 illustrates the TPA pyrolysis results obtained under the N_2 and CO_2 conditions. As shown in Fig. 10(a), CO formation was enhanced in the presence of CO_2, while the CO_2 concentration decreased, which is consistent with the trend observed in Fig. 8. Among the condensable products (Fig. 10(b)), all the identified condensable compounds, except phenol, decreased in mass under CO_2, indicating that CO_2 increased the thermal degradation rate. The exception of phenol might have been attributed to its relatively high thermal stability or secondary formation via the decomposition of oxygenated intermediates [58]. Notably, stable aromatic HCs, such as benzene, benzophenone, and biphenyl, were suppressed under CO_2 conditions, suggesting that CO_2 may facilitate secondary reactions beyond thermal degradation. These may include the ring-opening of aromatic intermediates (*e.g.*, benzene and biphenyl), and the decarboxylation or decarbonylation of aromatic ketones (*e.g.*, benzophenone).

Mechanistically, prior studies have suggested that CO_2 can be activated on the Ni surface via single- or two-electron transfer, potentially forming CO_2-derived surface species such as the CO_2 radical anion ($CO_2{\bullet}^-$) and/or metal-bound carboxylate/carbonate-type species (M-COO^-) [59,60]. $CO_2{\bullet}^-$ species, possessing electrophilic and radical

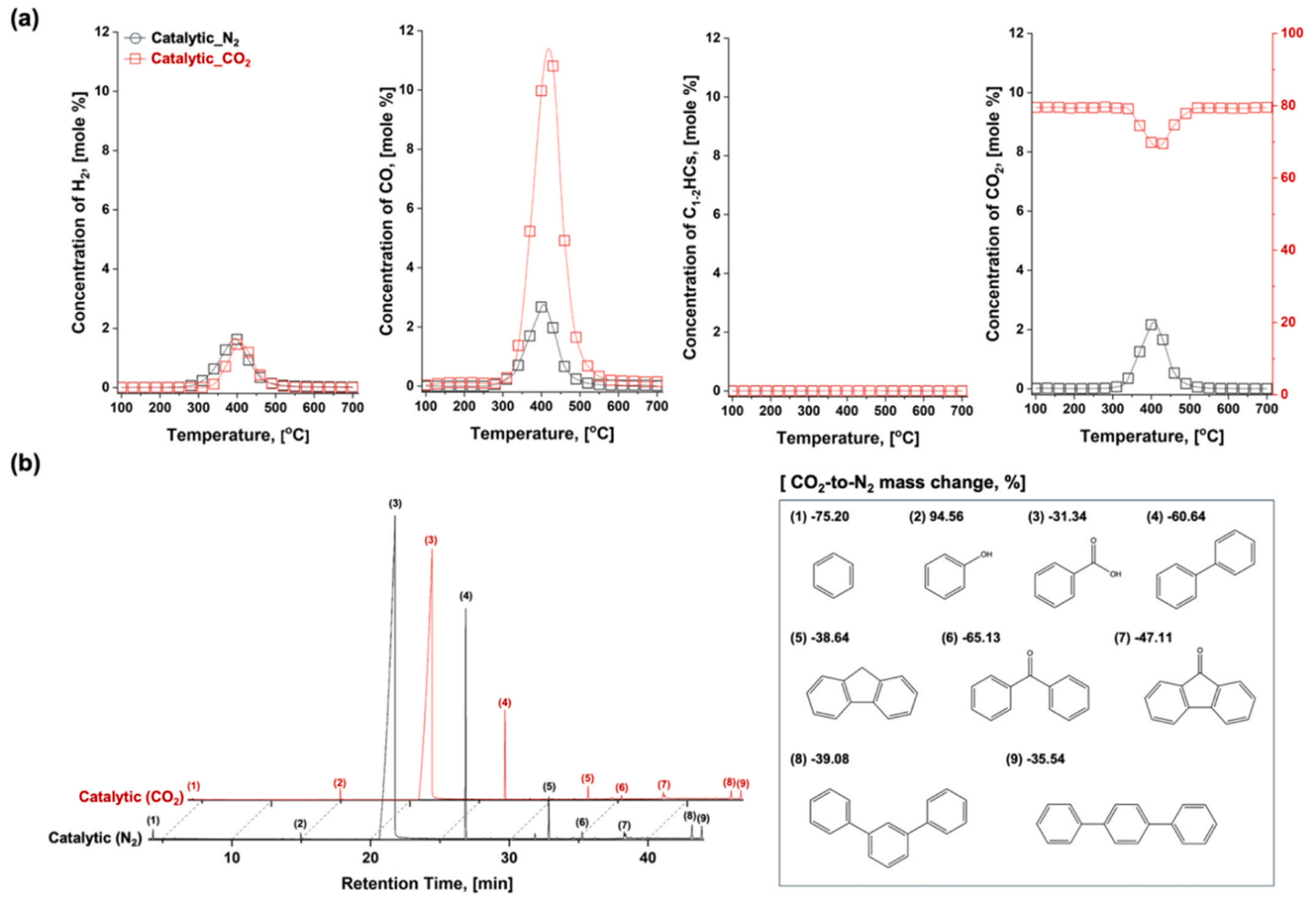

Fig. 10. (a) Concentration profiles of major gaseous products (H_2, CO, C_{1-2}HCs, and CO_2) that evolved from the catalytic pyrolysis of TPA over a Ni-based catalyst under N_2 (black squares) and CO_2 (red squares) conditions, and (b) chromatograms of condensable products obtained from catalytic pyrolysis under same conditions, with the identified compounds and their relative mass changes under CO_2 conditions, compared to those under N_2 conditions.

characteristics, have been proposed to interact with aromatic structures and/or facilitate the cleavage of bonds (C–C, C–H, and C–O), thereby promoting ring-opening reactions and enhancing CO evolution [61,62]. Additionally, the asymmetric structure of M-COO$^-$ may promote bond cleavage to form CO [60]. Collectively, the literature indicates that CO_2 may play an active role in Ni-catalyzed pyrolysis by participating in deoxygenation pathways (*e.g.*, decarboxylation and decarbonylation) and destabilizing aromatic structures. While CO_2 derived species such as $CO_2\bullet^-$ and M-COO$^-$ have been previously discussed in theoretical studies and in simple model systems, their specific roles in the catalytic pyrolysis of oxygenated aromatic compounds have not been clearly demonstrated. In this study, the mechanistic interpretation was directly supported by the unique gaseous and condensable product shifts observed during TPA pyrolysis under CO_2 conditions. These results indicate that, under catalytic conditions, reactive species generated from CO_2 may help break certain bonds and promote the oxidation of aromatic compounds such as TPA. Based on these mechanistic insights, a schematic representation of the proposed nickel-catalyzed CO_2-assisted degradation mechanism of PBT is provided in Fig. 11. Nevertheless, deeper identification of CO_2 activation routes and the interactions between CO_2-derived reactive species and PBT-derived fragments remains needed. Future work will therefore focus on elucidating these elementary steps via in-situ/operando techniques (*e.g.*, DRIFTS, XAS), exploring additional parameters that govern product selectivity, and evaluating the long-term stability of Ni-based catalysts under CO_2 conditions.

Building upon these mechanistic insights, it is also important to consider the long-term stability of Ni-based catalysts under CO_2 conditions. While Ni-based catalysts can suffer from deactivation via sintering

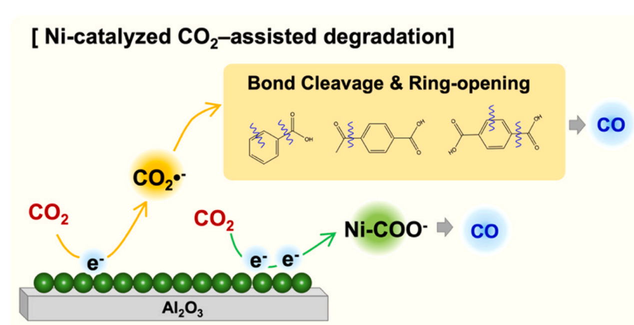

Fig. 11. Proposed Ni-catalyzed CO_2-assisted degradation mechanism.

or coke formation [63,64], our study focused on carbon deposition as a critical deactivation factor. To evaluate the catalyst stability, the extent of coke formation was evaluated using temperature-programmed oxidation (TPO) for two used catalysts exposed to N_2 and CO_2 (Fig. 12). The average coking rate, calculated based on the TGA mass loss, revealed a substantially higher coke yield under N_2 (3.83 wt%) compared to that under CO_2 (1.23 wt%). This indicated that CO_2 effectively mitigated catalyst deactivation. Importantly, the TPO profiles demonstrated that the deposited carbonaceous species could be completely removed via oxidation. Considering that carbon deposition primarily originated from the condensable products, this complete removal suggests that Ni catalyst possesses high regeneration potential,

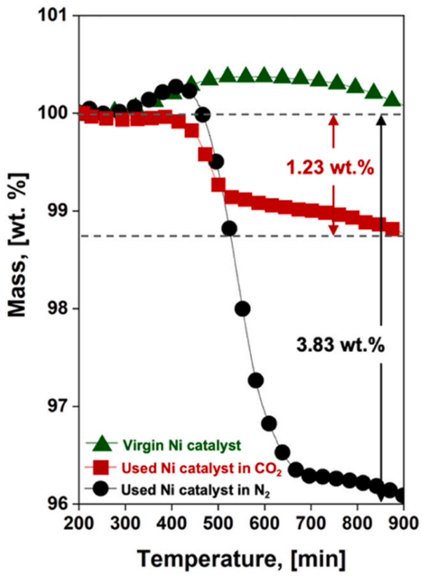

Fig. 12. TPO profiles of virgin and used Ni-based catalysts under N_2 and CO_2 conditions.

as the active sites blocked by coke can be effectively restored through oxidative treatment. Therefore, it is reasonable to infer that, under CO_2 conditions, a large fraction of these intermediates was diverted to CO formation rather than condensing and polymerizing on the catalyst surface. Furthermore, comparative analysis (SEM-EDS, H_2-TPR, and XRD) confirmed that the physicochemical properties of the spent catalyst, including crystal structure and reducibility, were well-preserved without significant sintering of phase transformation (Fig. S8).

In addition to the mechanistic advantages, the catalytic pyrolysis of PBT under CO_2 conditions offers environmental benefits. To evaluate the potential environmental relevance of CO_2-assisted catalytic pyrolysis, we quantified a reactor-boundary CO_2 balance by calculating the net moles of CO_2 consumed (CO_2 in - CO_2 out, based on the measured outlet gas composition; Fig. 13). By scaling this experimentally measured net CO_2 consumption from a 2 g PBT basis to the projected global PBT generation in 2030 (1.8 million tonnes) [65], we estimate a theoretical maximum CO_2 consumption potential on the order of approximately 6 million tonnes of CO_2. We emphasize that this value represents an upper-bound, reactor-boundary estimate and should not be interpreted as avoided emissions from a full life-cycle assessment. If this entire volume were disposed of via conventional incineration, it would result in the emission of nearly 4 million tonnes of CO_2 [66]. In comparison, pyrolysis under N_2 conditions, regardless of the configuration (one-stage, two-stage, or catalytic pyrolysis), resulted in minimal CO_2 mitigation, yielding near-neutral or slightly positive CO_2 emission potentials within the same reactor-boundary accounting. Under CO_2-assisted catalytic pyrolysis, the apparent benefit arises because CO_2 is used as a reactive agent and is partially converted into CO, thereby being incorporated into syngas rather than exiting the reactor as CO_2. Consequently, the process has the potential to both valorize waste PBT and utilize CO_2 as a feedstock. However, the net climate benefit at scale will depend on practical factors such as the fraction of PBT that is collected and processed, the achievable net CO_2 conversion under industrial operation, and indirect emissions associated with CO_2 supply and process energy. Therefore, while CO_2-assisted catalytic pyrolysis is a promising strategy for circular utilization of plastic waste and CO_2, a comprehensive LCA/TEA incorporating these factors and the downstream fate of CO/syngas is required to quantify realistic, system-level CO_2 mitigation.

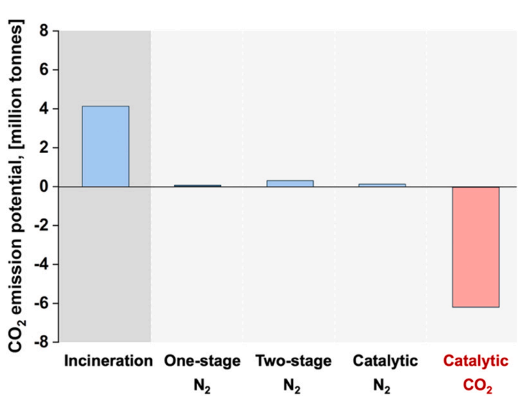

Fig. 13. Potential for CO_2 emissions from incineration, non-catalytic pyrolysis (one-stage and two-stage pyrolysis) under N_2 conditions, and catalytic pyrolysis under N_2 and CO_2 conditions.

4. Conclusions

This study compared PBT waste pyrolysis under three configurations (one-stage, two-stage, and catalytic pyrolysis) to maximize syngas recovery and to clarify the role of CO_2 as a reactive medium. Under N_2 one-stage pyrolysis, condensable products dominated (67.91 wt%) and syngas production was limited, whereas CO_2-assisted catalytic pyrolysis markedly redirected products toward gas: the condensable yield decreased to 10.28 wt% and CO became the major component, with CO formation increasing by ~65 times relative to N_2 one-stage operation. Mechanistically, the results support that CO_2 is activated on the Ni catalyst surface to generate reactive species such as $CO_2\bullet^-$ or M-COO^-, which facilitate selective C–C, C–H, and C–O bond cleavage, aromatic ring-opening, and deoxygenation, thereby suppressing heavy oxygenates and tar precursors. Although coke formation remained unavoidable, introducing CO_2 mitigated coking and reduced catalyst deactivation compared with N_2. Overall, integrating CO_2 with Ni-catalyzed pyrolysis provides a dual-benefit route that intensifies syngas production while enabling in-situ CO_2 utilization for ELV plastic valorization. However, several limitations must be acknowledged, particularly regarding the specific waste matric used in our experiments, which may influence the universality of these results. To address these limitations, further studies are warranted to investigate the effects of actual contaminants in PBT waste on the pyrolysis process. Additionally, the influence of experimental conditions, such as temperature and catalyst characteristics, should be systematically evaluated to better understand their effects on product yield and quality.

Accordingly, the syngas quality indicator H_2/CO decreased under CO_2 due to pronounced CO enrichment. From a deployment perspective, however, the product gas may still contain diluents and impurities whose allowable levels and removal requirements depend on the targeted downstream process. Therefore, a detailed quantification of impurities, application-specific upgrading/separation design, and an integrated techno-economic analysis (TEA) are warranted in future work to assess overall feasibility. Future work should also focus on improving catalyst stability (coke resistance and sustained CO_2 activation) and optimizing operating parameters (*e.g.*, CO_2-to-feed ratio, residence time), and conducting catalyst recycling studies to evaluate long-term performance. Additionally, we plan to investigate the multivariate influences of temperature gradients, CO_2 concentration, catalyst dosage, and heating rates. These future studies aim to clarify the optimal reaction conditions and enhance the practical applicability of our findings in industrial settings.

CRediT authorship contribution statement

Dohee Kwon: Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Conceptualization. **Jechan Lee:** Writing – review & editing, Methodology. **Eilhann E. Kwon:**

Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization. **Doyeon Lee:** Writing – review & editing, Resources. **Youngju Kim:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Jee Young Kim:** Writing – review & editing, Visualization, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Research Foundation of Korea (NRF), funded by the Korean Government (MSIT) (grant number No. RS-2023–00219667). Dohee Kwon also acknowledges financial support from the Hyundai Motor Chung Mong-Koo Foundation.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jaap.2026.107733.

Data availability

Data will be made available on request.

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